

मध्यावदी परीक्षा (Mid Semester Examination)

Name : _____ Adm. No. : _____

Subject : _____ Part : _____ Semester : _____

Signature of Invigilator : _____ Date : _____

Question No.	1	2	3	4	5	6	7	8	9	10	TOTAL
Marks Obtained											

Total Marks Obtained : _____ Signature of Examiner : _____

Ques-1(a) In a series of triaxial compression test on a Sandstone, the following represent the stresses at peak load conditions.

Test.	σ_3 (MPa)	σ_1 (MPa)
1	1.0	9.2
2	5.0	28.0
3	9.5	48.7
4	15.0	74.0

$$(a) \sigma_1 = 1.18 \text{ MPa}$$

$$\phi = 40^\circ$$

$$(b) p_w = 3.22 \text{ MPa}$$

$$(c) K = 0.217$$

$$(d) \nu = 0.173$$

Determine the value of C & ϕ that best fit the data.

(b) The initial state of stress at a point in the ground, in the same Sandstone is.

$$\sigma_3 = 8.96 \text{ MPa}$$

$$\sigma_1 = 34.47 \text{ MPa}$$

By the construction of a dam reservoir, the pore pressure (p_w) in the ~~dam~~ rock will increase.

What value of p_w will cause fracture of the Sandstone in-situ. (Assume the fracture occurs at peak stress).

(c) What is the value of the principal stress ratio

$\left(\frac{\sigma_3}{\sigma_1} \right)$ above which failure cannot occur?

(d) Imagine a case of dynamic loading of the said rock. If in a plane wave front, the pressure in the plane of the wave front are $\left(\frac{2}{1-\nu}\right)$ times the pressure normal to the wave front. What is the least value of Poisson's ratio (ν) such that compressive or shear failure does not occur as the wave front sweeps through the rock?

$$\begin{aligned} \sigma_1 &= 1.18 \text{ MPa} \\ \sigma_2 &= \phi \\ \sigma_3 &= 0.33 \text{ MPa} \\ \tau &= 0.15 \text{ MPa} \\ \nu &= 0.15 \end{aligned}$$

Test	σ_1 (MPa)	σ_2 (MPa)	σ_3 (MPa)
1	0.1	0.1	0.1
2	0.2	0.2	0.2
3	0.2	0.2	0.2
4	0.2	0.2	0.2

Determine the value of ν that best fits the data.

(b) The initial state of stress at a point in the ground is:

$$\sigma_1 = 2.0 \text{ MPa}$$

$$\sigma_2 = 3.0 \text{ MPa}$$

Estimate the construction of a dam reservoir to be placed in the rock with increase in the value of ν will cause fracture of the rock to occur. Assume the fracture occurs at point 2 (a).

(c) What is the value of the principal stress ratio $\left(\frac{\sigma_3}{\sigma_1}\right)$ above which failure cannot occur?

Solution-1

Given	σ_3 (MPa)	σ_1 (MPa)		σ_3^2	$\sigma_3 \cdot \sigma_1$
1	1.0	9.2	calculate →	1	9.2
2	5.0	28		25	140
3	9.5	48.7		90.25	462.65
4	15	74		225	1110
Sum	30.5	159.9		341.25	1721.85

$$\begin{pmatrix} m \\ b \end{pmatrix} = \begin{pmatrix} \sum \sigma_{3i}^2 & \sum \sigma_{3i} \\ \sum \sigma_{3i} & n \end{pmatrix}^{-1} \begin{pmatrix} \sum \sigma_{3i} \sigma_{1i} \\ \sum \sigma_{1i} \end{pmatrix}$$

$$= \begin{pmatrix} 341.25 & 30.5 \\ 30.5 & 4 \end{pmatrix}^{-1} \begin{pmatrix} 1721.85 \\ 159.9 \end{pmatrix}$$

$$\begin{pmatrix} m \\ b \end{pmatrix} = \begin{pmatrix} 4.624 \\ 4.714 \end{pmatrix}$$

Since

$$\sigma_1 = \frac{2c \cos \phi}{1 - \sin \phi} + \sigma_3 \left(\frac{1 + \sin \phi}{1 - \sin \phi} \right)$$

$$\text{or } \sigma_1 = b + m \sigma_3$$

$$m = 4.624 = \frac{1 + \sin \phi}{1 - \sin \phi}$$

$$\text{or, } \sin \phi = \left(\frac{m-1}{m+1} \right) = \left(\frac{4.624-1}{4.624+1} \right) = \frac{4.53}{7.03}$$

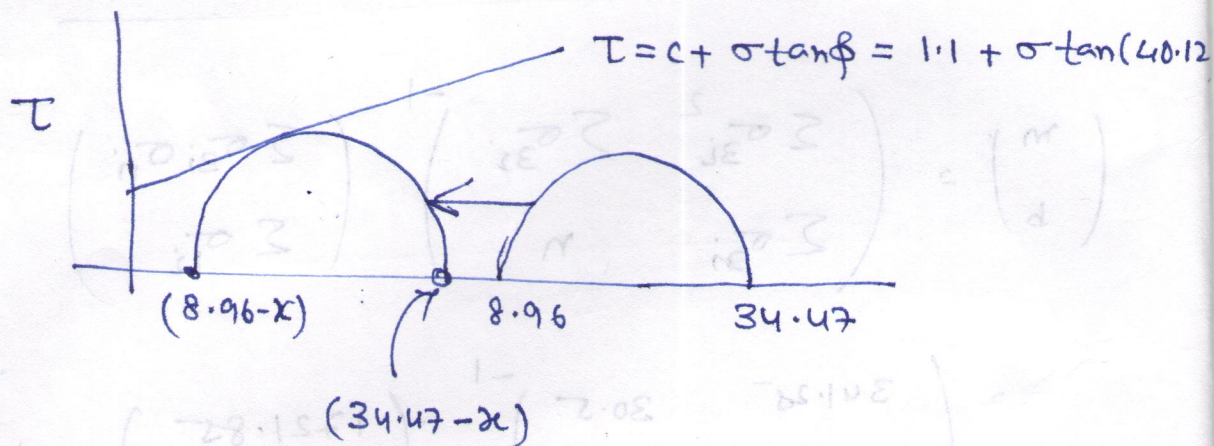
$$\Rightarrow \boxed{\phi = 40.12^\circ}$$

$$\frac{2c \cos \phi}{1 - \sin \phi} = b = 4.714 \text{ k}$$

$$\Rightarrow c = \frac{4.714 (1 - \sin 40.12)}{2 \cos (40.12)} \text{ MPa.}$$

$$= 1.09 \text{ MPa} \approx \underline{1.1 \text{ MPa}}$$

(b)



The presence of pore water by x MPa is shifting the Mohr Circle to the left by x units.

This Mohr circle is going to touch, if failure is to occur.

Hence, the ^{principal} stresses in the Mohr circle are going to satisfy the failure criteria.

$$\text{ie, } (\sigma_1, \sigma_3) = (34.47 - x, 8.96 - x)$$

will satisfy the failure criteria. given by

$$\sigma_1 = b + m \sigma_3.$$

$$\text{Or, } (34.47 - x) = b + m(8.96 - x)$$

$$\text{Or, } = b + 8.96m - mx$$

$$\text{Or, } mx - x = (b + 8.96m - 34.47)$$

$$\text{Or, } x = \frac{b + 8.96m - 34.47}{(m - 1)}$$

In Part (a) we got $b = 4.714$

$$\text{Or } m = 4.624$$

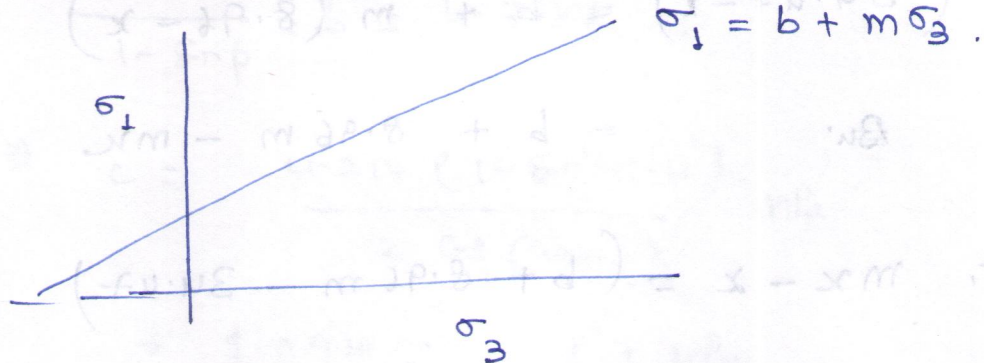
$$\therefore x = \frac{4.714 + 8.96 \times 4.624 - 34.47}{(4.624 - 1)} \text{ MPa.}$$

$$x = 3.22 \text{ MPa.}$$

Therefore, the Value of pore water pressure of

(3.22 MPa) is going to create fracture in the rock.

(C)



The failure criteria is given by

$$\sigma_1 = b + m \sigma_3 \quad \text{--- (Failure line)}$$

Dividing both sides by σ_1 .

$$1 = \frac{b}{\sigma_1} + m \left(\frac{\sigma_3}{\sigma_1} \right) \quad \text{--- (Failure line)}$$

$$\text{Let } \frac{\sigma_3}{\sigma_1} = R_x$$

$$\Rightarrow 1 = \frac{b}{\sigma_1} + R_x m$$

$$\text{or, } \frac{b}{\sigma_1} = 1 - R_x m \quad \text{--- (Failure line)}$$

$$\text{or, } \sigma_1 = \frac{b}{1 - R_x m} \quad \text{(Failure line)}$$

Now as $R_x \rightarrow \frac{1}{m}$

$\sigma_1 \rightarrow \infty$ (Implies that the rock cannot fracture because it requires infinite stress to fracture the rock).

Now if $R_x > \left(\frac{1}{m}\right)$

$\Rightarrow \sigma_1$ is (-ve) quantity.

But as per the experimental Configuration σ_1 is compressive. This is invalidating the failure Criterion,

In other words this value of $R_x > \frac{1}{m}$ is dragging the σ_1 to a negative region which is undefined as per failure Criteria & hence the failure is not going to take place.

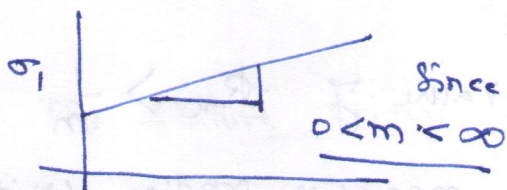
Hence the value of $\left(\frac{\sigma_3}{\sigma_1}\right) = \frac{1}{m} = \frac{1}{4.624} = 0.21$

is the minimum value at which the fracture is not going to take place.

In other words if $R_x < \frac{1}{m}$ The N_r is < 1
& the ratio is more than $\frac{\sigma_1}{\sigma_3}$ on the line
& Hence, brittle failure will take place.

(d) The case of dynamic loading shows that,

$$\sigma_{\text{plane of wave}} = \left(\frac{\nu}{1-\nu} \right) \sigma_{\text{normal to wave.}}$$



$$\sigma_3 \Rightarrow \sigma_1 > \sigma_3 \text{ or } \sigma_3 < \sigma_1$$

For brittle failure to occur.

$$\text{Since } \nu < 0.5 \Rightarrow 2\nu < 1 \text{ or } \nu < (1-\nu)$$

$$\Rightarrow \frac{\nu}{1-\nu} < 1$$

$$\frac{\sigma_{\text{plane of wave}}}{\sigma_{\text{Normal to wave}}} = \frac{\nu}{1-\nu} \text{ is as good as saying}$$

$$\left(\frac{\sigma_3}{\sigma_1} \right) = \left(\frac{\nu}{1-\nu} \right)$$

But from last ~~ex~~ problem part (c).

$$\frac{\sigma_3}{\sigma_1} = k_x = 0.21$$

$$\therefore 0.21 = \frac{\nu}{1-\nu}$$

$$\text{or } \nu = 0.21 - 0.21\nu$$

$$\text{or } \nu = \left(\frac{0.21}{1+0.21} \right) = 0.173$$

$\therefore$ The minimum value of ν at which failure (Brittle / compressive) does not occur is

$$\boxed{\nu = 0.173}$$